

Bank ALM Master Model

Major European G-SIB — Asset-Liability Management Framework

Built to institutional standard | Calibrated to Q4 2025 public disclosures | v10

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Role	ALM, IRRBB & Liquidity Risk Intern
Data	Major European G-SIB — Q4 2025 Actuals (Public Disclosure)
Platform	Microsoft Excel (.xlsx) 1,000+ Formulas 0 Errors
Architecture	18 Analytical Sections 4 Scenarios 6 Sheets
Regulatory	EBA/GL/2022/14 CRR2 / Basel III Del. Reg. (EU) 2022/786
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Executive Summary

Bank ALM Master Model — Major European G-SIB | Hossam Eltarrass, ALM, IRRBB & Liquidity Risk Intern | May 2026

This showcase presents an institutional-grade Asset-Liability Management (ALM) model calibrated to Q4 2025 public disclosures of a major European G-SIB. The model spans 18 analytical sections across 6 workbook sheets, delivering a complete view of interest rate risk, liquidity risk, and capital adequacy under four rate scenarios — fully aligned with EBA/GL/2022/14 IRRBB standards and the Basel III CRR2 capital framework. A single-source-of-truth architecture (one Engine scenario selector cascading via CHOOSE formulas) ensures consistency across 1,000+ linked formulas with zero errors.

Key Results — 2025A Actuals (Base)

€22.09 bn	1.73%	15.75%	155.4%	102.0%	13.2%
NII	NIM	CET1	LCR	NSFR	EVE/T1
2025A Actuals	2025A Actuals	min 13.5% PASS	min 100% PASS	min 100% PASS	outlier >15% PASS

Analytical Coverage

Domain	Sections Covered	Regulatory Basis
Interest Rate Risk	NII simulation, IRRBB (6 EBA shocks), PVBP/DV01, EVE cashflow discounting (Pillar 3), repricing gap	EBA/GL/2022/14
Liquidity Risk	LCR, NSFR, maturity ladder (EBA/GL/2014/35), monthly NII forward schedule	Del. Reg. (EU) 2022/786; CRR Art.428a-428at; EBA/GL/2014/35
Capital Adequacy	CET1, leverage ratio, MDA buffer trajectory	CRR2 / Basel III Art.92
Profitability	Full P&L, NIM trend (2024A–2028F), NIM sensitivity, FTP curve with worked examples	BCBS 239 / EBA internal governance
Stress Testing	ICAAP severe scenario, 4 rate scenarios, CPR model	EBA/GL/2022/14; EBA SREP
Hedge Effectiveness	IRS overlay, 55% hedge ratio, EVE post-hedge by shock	EBA/GL/2022/14 Annex II

Key Findings

- Asset-sensitive balance sheet: NII rises €7.7bn under +200bps parallel rate shock — NII sensitivity 34.7%, monitored per EBA/GL/2022/14 para 116
- All regulatory thresholds pass in 2025A: CET1 (15.75%), LCR (155.4%), NSFR (102.0%), EVE/T1 (13.2%), Leverage (4.73%)
- EVE/Tier 1 of 13.2% is below the 15% EBA outlier threshold — 55% IRS hedge offset quantified in Section 18
- CET1 trajectory: 15.75% (2025A) → 16.13% (2028F); MDA buffer expands from 2.25% to 2.63% over forecast period
- NII: €22.09bn (2025A) → €24.30bn (2028F) driven by 3% annual loan growth and 40% deposit beta (base scenario)
- ICAAP severe scenario: Net income remains positive at €2.2bn; CET1 holds at 16.49% under stress (Section 17)
- FTP curve upward-sloping across all tenors: 5Y mortgage FTP charge 3.21%; overnight NMD FTP credit 2.25% (Section 15)
- CPR model (Section 16): Prepayment accelerates in rate-cut environment — effective loan duration shortens, reducing EVE exposure

Data Source & Disclaimer: Balance sheet actuals calibrated to Q4 2025 public disclosures of a major European G-SIB (publicly available Pillar 3 report). No proprietary or confidential data used. All forecast years (2026F–2028F) are model-generated. Regulatory references are to publicly available EBA guidelines and EU regulations. Built for portfolio purposes only.

Model Architecture

Six-sheet workbook — single-source-of-truth design, zero independent dropdowns

18	1,000+	0	4	6
Analytical Sections	Excel Formulas	Formula Errors	Stress Scenarios	Workbook Sheets

Workbook Architecture

Sheet	Role	Key Contents
Cover	Summary	Model overview, key metrics snapshot, author
Dashboard	ALCO View	KPI cards, balance sheet forecast, IRRBB summary, scenario comparison
Master Bal. Sheet	Source of Truth	2024A–2025A actuals; 3-year forecast; HQLA flags; CSV export for Python
Engine	Control Layer	Rate curve, behavioural parameters — CHOOSE formula propagation
YieldCurve	Yield Curve	Zero-coupon bootstrap; forward rates; feeds EVE discounting (S.14b)
Model	Analytical Engine	18 sections: NII, IRRBB, LCR, NSFR, CET1, PVBP, FTP, CPR, ICAAP, hedge

18 Analytical Sections

#	Section	Regulatory Reference
1	NII Simulation	EBA/GL/2022/14
2	P&L Model (2025A–2028F)	—
3	Static Repricing Gap	EBA/GL/2022/14
4	IRRBB — 6 EBA Shock Scenarios (NII + EVE)	EBA/GL/2022/14 Annex I
5	LCR	Delegated Regulation (EU) 2022/786
6	NSFR	CRR Art. 428a–428at
7	Capital Adequacy (CET1, Leverage, MDA)	CRR2 / Basel III Art. 92
8	NIM Trend (2024A–2028F)	—
9	Forecast Regulatory Ratios — Traffic Light	—
10	Scenario Comparison (Base / Hike / Cut / Stress)	—
11	NIM Sensitivity (ECB Rate vs Deposit Beta)	EBA/GL/2022/14 para 116
12	Monthly NII Forward Schedule (12-month)	EBA/GL/2022/14
13	Maturity Ladder — Liquidity Stress	EBA/GL/2014/35
14	PVBP / DV01 — Duration-Weighted EVE	EBA/GL/2022/14 Annex II
14b	EVE Full Cashflow Discounting — Pillar 3 Method	CRR Art. 448 / EBA/GL/2022/14 Annex II
15	FTP Curve — Internal ALM Pricing	BCBS 239 / EBA governance
16	CPR / Prepayment Sensitivity	EBA/GL/2022/14 Annex II
17	ICAAP Stress Summary — Severe Scenario	EBA SREP guidelines
18	Hedge Effectiveness — IRS Overlay	EBA/GL/2022/14

Scenario Framework — Single Dropdown Control (Engine!C5)

Scenario	Rate Shock	Loan Growth	Dep. Growth	Dep. Beta	CoR (bps)	CIR
1 — Base	0 bps	3.0%	2.5%	0.40	25	62%
2 — Rate Hike	+200 bps	2.0%	2.0%	0.50	30	63%
3 — Rate Cut	-200 bps	4.0%	3.0%	0.30	20	61%
4 — Stress	+300 bps	1.0%	0.0%	0.60	60	65%

Key Design Decisions

Single-source-of-truth: Engine!C5 drives all 18 sections via CHOOSE — no independent dropdowns anywhere

YieldCurve sheet: Bootstrapped zero-coupon curve feeds Section 14b cashflow discounting; reconciled against PVBP method

EVE hedge offset: 55% IRS overlay (Section 18) — unhedged EVE ~29% misrepresents actual risk posture

CPR model: Prepayment sensitivity calibrated across 4 scenarios; duration scaling per EBA/GL/2022/14 Annex II

ICAAP stress: Independent NII, credit, and behavioural overlays — ALCO-ready traffic-light output (Section 17)

Circular reference: Cash plug dependency resolved using prior-year cash balance

Colour convention: Blue = hardcoded input | Black = formula | Green = cross-sheet reference

Dashboard — Executive ALCO View

Major European G-SIB | EUR | Scenario: Base | Forecast 2026F–2028F | Basel III / CRR2 / Del. Reg. (EU) 2022/786

€22.69 bn	1.74%	15.88%	153.5%	102.0%	13.2%
NII	NIM	CET1	LCR	NSFR	EVE/T1
+€600m vs 2025A	+0.01pp vs 2025A	min 13.5% PASS	min 100% PASS	2025A PASS	outlier >15% PASS

Balance Sheet Forecast (EUR millions)

Line Item	2025A	2026F	2027F	2028F
Cash & CB Reserves	211,330	214,325	217,158	219,876
Trading Assets	870,245	870,245	870,245	870,245
Loans to Customers	897,358	924,279	952,007	980,567
Interbank Loans	26,259	26,259	26,259	26,259
Securities	153,107	157,700	162,431	167,304
Other Assets	634,682	641,029	647,439	653,913
TOTAL ASSETS	2,792,981	2,833,837	2,875,539	2,918,165
Customer Deposits	1,075,564	1,102,453	1,130,014	1,158,265
Interbank + Repo Funding	427,885	427,885	427,885	427,885
Bonds Issued + Sub Debt	208,401	208,401	208,401	208,401
Other Liabilities	948,958	958,448	968,032	977,712
Equity	132,173	136,650	141,207	145,902
TOTAL L + E	2,792,981	2,833,837	2,875,539	2,918,165

Profitability (EUR millions)

P&L Line	2025A	2026F	2027F	2028F
Net Interest Income	22,085	22,686	23,225	24,296
Fee Income	22,344	22,671	23,004	23,345
Total Revenue	44,429	45,356	46,229	47,641
Operating Expenses	-27,546	-28,121	-28,662	-29,538
Provisions / CoR	-2,243	-2,311	-2,380	-2,451
Net Income	10,980	11,194	11,390	11,739

Key Ratios

Ratio	Threshold	2025A	2026F	2027F	2028F
NIM	—	1.73%	1.74%	1.73%	1.77%
ROE	—	8.31%	8.33%	8.06%	8.18%
Cost-to-Income	—	62.0%	62.0%	62.0%	62.0%
CET1 Ratio	>= 13.5%	15.75%	15.88%	16.00%	16.13%
LCR	>= 100%	155.4%	153.5%	154.7%	155.8%
NSFR	>= 1.00x	1.020x	0.960x	0.960x	0.959x
Leverage Ratio		4.73%	4.82%	4.91%	5.00%

NSFR 2025A: 102.0% (PASS). Forecast NSFR 96.0% — below 1.00x threshold. Forecast decline driven by repo RSF treatment under v10 — identified and documented — methodology assumption, not a balance sheet risk. 2025A: 102.0% (PASS).

Scenario Comparison — 2026F

Metric	Base	Rate Hike	Rate Cut	Stress
NII (€bn)	22.69	20.22	18.82	15.81
NIM	1.74%	1.55%	1.44%	1.21%
Net Income (€bn)	11.19	9.82	10.75	5.94
CET1	15.88%	15.90%	15.78%	16.02%
EVE / Tier 1	13.2%	13.2%	11.3%	19.9%*

* EVE/T1 under Stress +300bps = 19.9% — breaches 15% outlier threshold. Gross unhedged reading; hedge overlay (Section 18) reduces to ~10.9%.

Master Balance Sheet — Source of Truth

Granular G-SIB balance sheet | 2024A–2025A actuals from Q4 public disclosures | 3-year forecast | HQLA eligibility flags

Assets (EUR millions)

Line Item	2024A	2025A	HQLA
Cash and balances at central banks	182,496	211,330	Level 1
Financial instruments FVTPL — trading	267,357	321,293	—
Financial instruments FVTPL — non-trading	225,699	254,310	—
Financial instruments FVTPL — designated at FV	322,631	274,625	—
Derivatives used for hedging (asset)	20,851	20,017	—
Financial assets FVOCI — equity instruments	71,430	77,940	—
Financial assets FVOCI — debt instruments	1,610	1,420	—
Loans & receivables at AC — interbank	31,147	26,259	—
Loans & receivables at AC — customers	900,141	897,358	—
Debt securities at amortised cost	146,975	151,687	Level 1 / 2A
Remeasurement adj. on IR-hedged portfolios	-758	-2,335	—
Insurance investment assets	286,849	305,471	—
Tax assets, accrued income, other	180,362	173,534	—
Equity method investments	7,862	6,950	—
Property, plant, equipment & intangibles	54,706	58,184	—
TOTAL ASSETS	2,670,387	2,792,981	—

Liabilities & Equity (EUR millions)

Line Item	2024A	2025A
Central bank deposits	2,753	4,401
Financial instruments FVTPL — trading	261,178	294,642
Financial instruments FVTPL — designated	274,703	255,312
Derivatives used for hedging (liability)	17,759	20,051
Customer deposits	991,875	1,075,564
Debt securities issued	163,506	173,933
Subordinated debt	25,804	34,468
Interbank borrowings	67,437	69,938
Repo funding	282,789	357,947
Other liabilities	454,069	374,551
Total Equity	128,514	132,173
TOTAL LIABILITIES + EQUITY	2,670,387	2,792,981

Data source: Q4 2025 public disclosures of a major European G-SIB (Pillar 3 report). HQLA classification per Delegated Regulation (EU) 2022/786. Forecast years (2026F–2028F) generated by Engine growth assumptions. Master Balance Sheet structured for CSV export and Python pipeline consumption.

Engine — Central Control Layer

Rate curve, scenario inputs, spread & behavioural assumptions | All sections reference Engine via CHOOSE formulas

Section A1 — Scenario Control

Engine!C5 controls the active scenario (Data Validation: 1–4). All downstream CHOOSE formulas reference this one cell — changing it cascades simultaneously through NII, EVE, LCR, NSFR, and CET1.

Scenario	Rate Shock	Loan Growth	Dep. Growth	Dep. Beta	CoR (bps)	CIR
1 — Base	0 bps	3.0%	2.5%	0.40	25	62%
2 — Rate Hike	+200 bps	2.0%	2.0%	0.50	30	63%
3 — Rate Cut	-200 bps	4.0%	3.0%	0.30	20	61%
4 — Stress	+300 bps	1.0%	0.0%	0.60	60	65%

Section A2 — Market Rate Environment (ECB / Bloomberg)

Parameter	2024A	2025A	2026F	2027F	2028F
ECB Policy Rate (DFR)	4.00%	2.50%	2.25%	2.00%	2.00%
3M Euribor	3.90%	2.60%	2.35%	2.10%	2.10%
5Y Swap Rate	2.90%	2.80%	2.70%	2.60%	2.55%
10Y Swap Rate	2.70%	2.60%	2.55%	2.50%	2.45%

Section A3 — Spread & Behavioural Assumptions

Parameter	2025A	2026F+	Source
Loan spread over 3M Euribor	+120 bps	+120 bps	G-SIB loan yield minus Euribor avg.
Deposit beta	40%	Scenario	Scenario-dependent (Base: 0.40)
Repo spread below ECB DFR	-8.5 bps	-8.5 bps	ECB DFR minus ESTR (~8.5bps)
Senior unsecured bond spread over 10Y	+65 bps	+65 bps	G-SIB SU credit spread
Sub debt blended cost (AT1+T2)	2.50%	2.50%	Blended AT1+T2 G-SIB Pillar 3
Retail dep. behavioural duration	2.5 yrs	2.5 yrs	EBA/GL/2022/14 Table 2
NMD core % (stable)	70%	70%	BCBS IRRBB 2016 para 114
Tax rate	25%	25%	Effective rate
Dividend payout	60%	60%	G-SIB stated policy

Section A4 — NII Rate Mapping (Scenario-Dependent)

Asset / Liability	Rate Anchor	Base	Rate Hike	Rate Cut	Stress
Cash & CB reserves	ECB DFR	2.50%	4.50%	0.50%	5.50%
Loans to customers	Euribor + 120bps	3.75%	5.75%	1.75%	6.75%
Interbank loans	Euribor + 50bps	3.10%	5.10%	1.10%	6.10%
Securities	5Y Swap + 40bps	3.10%	5.10%	1.10%	6.10%
Customer deposits	DFR x Beta	0.50%	1.38%	0.00%	2.15%
Repo funding	DFR - 8.5bps	2.42%	4.42%	0.42%	5.42%

Model — NII Simulation & P&L;

Section 1: NII by asset class | Section 2: P&L; 2025A–2028F | Section 12: Monthly NII forward schedule

Section 1 — NII Simulation 2025A (Base Scenario)

Asset / Liability	Balance (EUR m)	Rate	Interest (EUR m)
Cash & CB Reserves (earns ECB DFR)	211,330	2.50%	5,283
Loans to Customers	897,358	3.80%	34,100
Interbank Loans	26,259	3.10%	814
Securities Portfolio	153,107	3.00%	4,593
TOTAL INTEREST INCOME	1,288,054	—	44,790
Customer Deposits	1,075,564	0.50%	5,378
Interbank Funding	69,938	3.10%	2,168
Repo Funding	357,947	2.42%	8,644
Bonds Issued	173,933	3.25%	5,653
Subordinated Debt	34,468	2.50%	862
TOTAL INTEREST EXPENSE	1,711,850	—	22,705
NET INTEREST INCOME	—	—	22,085

Section 2 — P&L; Model (2025A–2028F, Base Scenario)

P&L Line	2025A	2026F	2027F	2028F
Net Interest Income	22,085	22,686	23,225	24,296
Fee Income	22,344	22,671	23,004	23,345
Total Revenue	44,429	45,356	46,229	47,641
Operating Expenses	-27,546	-28,121	-28,662	-29,538
Gross Operating Income	16,883	17,235	17,567	18,104
Provisions / CoR	-2,243	-2,311	-2,380	-2,451
Pre-Tax Income	14,640	14,925	15,187	15,652
Income Tax (25%)	-3,660	-3,731	-3,797	-3,913
Net Income	10,980	11,194	11,390	11,739
Retained Earnings	4,392	4,477	4,556	4,696

Section 12 — Monthly NII Forward Schedule (2026F, First 6 Months)

Month	Loan Bal (EUR m)	Loan Rate	Loan Inc.	Dep Bal (EUR m)	Dep Rate	Monthly NII
Jan	899,601	3.84%	2,879	1,077,805	0.52%	1,871
Feb	901,850	3.82%	2,871	1,080,050	0.51%	1,874
Mar	904,105	3.80%	2,863	1,082,300	0.50%	1,877
Apr	906,365	3.78%	2,855	1,084,555	0.49%	1,879
May	908,631	3.76%	2,846	1,086,815	0.48%	1,882
Jun	910,903	3.74%	2,838	1,089,079	0.47%	1,884

Full 12-month schedule in model. Annual NII reconciliation: monthly model €22,626m vs annual model €22,686m — €59m gap (<0.3%); structural scope gap documented.

Model — IRRBB, Repricing Gap & DV01

Section 3: Repricing gap | Section 4: IRRBB (6 EBA shocks) | Section 14/14b: PVBP & EVE cashflow discounting | EBA/GL/2022/14

Section 3 — Static Repricing Gap (2025A)

Time Bucket	Assets (EUR m)	Liabilities (EUR m)	GAP	Cum. GAP
Overnight	224,460	830,135	-605,676	-605,676
0–3 Months	828,407	103,265	+725,142	+119,466
3–6 Months	25,603	75,289	-49,687	+69,780
6–12 Months	42,231	112,934	-70,703	-923
1–2 Years	40,913	205,617	-164,704	-165,627
2–5 Years	71,535	312,835	-241,300	-406,927
5+ Years	54,906	179,331	-124,425	-531,352

Section 4 — IRRBB Shock Scenarios (EBA/GL/2022/14)

Shock Scenario	NII Impact (EUR m)	EVE Impact (EUR m)
Parallel Up +200bps	+7,665	-17,506
Parallel Down -200bps	-6,516	+14,880
Steeper	+3,449	-8,019
Flattener	-3,449	+8,019
Short Rate Up	+9,582	+197
Short Rate Down	-8,145	-167

EVE / Tier 1 Capital: 13.2% | Outlier threshold: <15% | Status: PASS

NII Sensitivity (+200bps / NII): 34.7% | Worst NII scenario: Short Rate Up (+€9,582m) | Monitored per EBA/GL/2022/14 para 116

Section 14b — EVE Full Cashflow Discounting (CRR Art.448 / Pillar 3)

Shock Scenario	Shocked EVE	Base EVE	ΔEVE (EUR m)	ΔEVE / T1	Status
Parallel Up +200bps	-441,587	-423,796	-17,791	13.5%	PASS
Parallel Down -200bps	-409,212	-423,796	+14,584	11.0%	PASS
Steeper	-431,815	-423,796	-8,019	6.1%	PASS
Flattener	-415,777	-423,796	+8,019	6.1%	PASS
Short Rate Up	-423,599	-423,796	+197	0.1%	PASS
Short Rate Down	-423,963	-423,796	-167	0.1%	PASS

Section 14 — PVBP / DV01 (Duration-Weighted EVE)

Time Bucket	Repricing GAP	Duration (yrs)	PVBP (EUR m/bp)	% of Total
Overnight	-605,676	0.003	-0.18	0.1%
0–3 Months	+725,142	0.125	+9.06	4.7%
3–6 Months	-49,687	0.375	-1.86	1.0%
6–12 Months	-70,703	0.750	-5.30	2.7%
1–2 Years	-164,704	1.500	-24.71	12.7%
2–5 Years	-241,300	3.500	-84.46	43.4%
5+ Years	-124,425	7.000	-87.10	44.8%

Total PVBP: -194.5 EUR m per 1bp | Net hedged EVE impact (+100bps): -8,753 EUR m | 55% IRS hedge offset (Section 18). PVBP method per EBA/GL/2022/14 Annex II.

Model — LCR, NSFR & Capital Adequacy

Section 5: LCR per Del. Reg. (EU) 2022/786 | Section 6: NSFR per CRR Art. 428a–428at | Section 7: CET1 per CRR2

Section 5 — LCR (Delegated Regulation (EU) 2022/786)

Component	2025A	2026F	2027F	2028F
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Level 1: Cash & CB Reserves (0%)	211,330	214,325	217,158	219,876
Level 1: Government Bonds (0%)	110,237	113,544	116,950	120,459
Level 2A: Corporate Bonds (15% haircut)	42,870	37,533	38,659	39,818
Total HQLA	364,437	365,402	372,767	380,154
Retail Deposits — stable (5%)	37,645	38,586	39,551	40,539
Retail Deposits — less stable (10%)	32,267	33,074	33,900	34,748
Wholesale non-operational (25%)	43,483	43,483	43,483	43,483
Interbank (financial inst.) (100%)	69,938	69,938	69,938	69,938
Undrawn commitments (10%)	89,736	92,428	95,201	98,057
Total 30-day Outflows	296,027	300,580	305,250	310,044
Total 30-day Inflows (capped at 75%)	61,573	62,607	64,306	66,045
Net 30-day Outflows	234,454	237,972	240,944	243,999
LCR RATIO	155.4%	153.5%	154.7%	155.8%

Section 6 — NSFR (CRR Art. 428a–428at) — 2025A Actuals

Component	Amount (EUR m)	Factor	ASF (EUR m)
Equity (CET1)	132,173	100%	132,173
Retail deposits — stable (Art.428h)	752,895	90%	677,605
Retail deposits — less stable (Art.428i)	322,669	80%	258,135
Bonds issued >1Y (Art.428n)	173,933	100%	173,933
Sub debt >1Y (Art.428o)	34,468	50%	17,234
Interbank < 6M (Art.428p)	69,938	0%	0
Total ASF	—	—	1,259,081
Loans to customers (85%/50% blend)	897,358	—	684,235
Gov bonds — Level 1 (5% RSF, Art.428r)	110,237	5%	5,512
Insurance assets (100% RSF)	305,471	100%	305,471
Trading — Sovereign/Level 1 (5% RSF)	348,098	5%	17,405
Total RSF	—	—	1,235,095
NSFR RATIO — 2025A	—	—	102.0%

Section 7 — Capital Adequacy (CRR2 / Basel III)

Metric	Threshold	2025A	2026F	2027F	2028F
Risk-Weighted Assets (EUR m)	—	839,428	860,538	882,280	904,675
CET1 Capital (EUR m)	—	132,173	136,650	141,207	145,902
CET1 Ratio	>= 13.5%	15.75%	15.88%	16.00%	16.13%
CET1 Buffer above MDA	—	2.25%	2.38%	2.50%	2.63%
Leverage Ratio	>= 3.0%	4.73%	4.82%	4.91%	5.00%

CET1 ratio 15.75% vs 13.5% MDA trigger — 225bps buffer. Leverage ratio 4.73% vs 3.0% threshold. RWA density 53.7% of total assets.

CET1 built up through retained earnings net of distributions; RWA growth tied to loan book expansion. SREP add-on modelled at aggregate level per CRR2 Art. 92.

Regulatory Summary — All Key Ratios (2025A)

Metric	Regulation	Threshold	2025A	Status
LCR	Del. Reg. (EU) 2022/786	>= 100%	155.4%	PASS
NSFR	CRR Art. 428a–428at	>= 100%	102.0%	PASS
CET1 Ratio	CRR2 Art. 92 / Basel III	>= 13.5%	15.75%	PASS
Leverage Ratio	CRR2 Art. 429	>= 3.0%	4.73%	PASS
EVE / Tier 1	EBA/GL/2022/14 para 98	< 15%	13.2%	PASS
NII Sensitivity	EBA/GL/2022/14 para 116	Monitored	34.7%	MONITOR

All prudential ratios in compliance as at 2025A. NSFR forecast divergence (2026F–2028F: ~96%) reflects repo RSF treatment under v10 — the methodology assumption was identified, stress-tested, and is identified, stress-tested, and documented. This is a documented model disclosure, not a balance sheet risk.

Python Extension Roadmap

Master Balance Sheet CSV is the primary data input for all Python tools | All 12 projects complete

This Excel model is the analytical foundation for a 12-project Python ALM toolkit — all 12 projects now complete. Each project consumes the Master Balance Sheet as a structured CSV and extends the Excel calculations, adding dynamic rate curves, interactive dashboards, and EBA-compliant stress testing logic. Full repository published on GitHub.

Python Projects — All 12 Complete

#	Project	Key Output	Model Dependency
1	NII Scenario Tool	NII across 4 rate scenarios; sensitivity table	Engine rates, NII Section
2	IRRBB Stress Testing	6 EBA shocks; EVE + NII outlier classification	IRRBB Section, EVE, NII
3	LCR / NSFR Calculator	HQLA stack; ASF/RSF table; ratio vs threshold	LCR & NSFR Sections
4	Bond Pricing & Duration	Clean/dirty price; modified duration; DV01	Securities, Yield Curve
5	Yield Curve Bootstrap	Zero-coupon curve; Nelson-Siegel fit; fwd rates	Engine rates, YieldCurve
6	FTP Framework	Tenor-matched FTP; business line NIM attribution	FTP Section (15)
7	CPR Behavioural Model	Prepayment curve by scenario; duration adjustment	CPR Section (16), IRRBB
8	ICAAP Stress Testing	Severe P&L; CET1 depletion; traffic-light output	ICAAP Section (17)
9	Hedge Effectiveness	IRS overlay; hedge ratio; EVE post-hedge by shock	Hedge Section (18)
10	Monthly NII + Maturity Ladder	12-month NII engine; cumulative liquidity gap	Sections 12 & 13
11	Capital Forecasting	RWA growth; CET1 trajectory; MDA buffer path	Capital Section (7)
12	ALM Dashboard Capstone	Dash app — all KPIs, multi-scenario toggle	Full integration

Regulatory Framework Referenced

Metric	Threshold	Regulatory Reference
LCR	>= 100%	CRR2 / Delegated Regulation (EU) 2022/786
NSFR	>= 100%	CRR Art. 428a–428at
IRRBB EVE	< 15% of Tier 1	EBA/GL/2022/14 — mandatory outlier test (Art.448 CRR)
IRRBB NII	>5% = EWI	EBA/GL/2022/14 para 116
CET1	>= 8% + buffers	CRR2 / Basel III Art. 92
Leverage	>= 3.0%	CRR2 Art. 92(1)(d)
FTP Curve	Internal	BCBS 239 / EBA internal governance guidelines
ICAAP	Severe scenario	EBA SREP guidelines
Maturity Ladder	—	EBA/GL/2014/35 — liquidity stress testing
Hedge Effectiveness	—	EBA/GL/2022/14 Annex II

Data Source & Disclaimer: Balance sheet actuals calibrated to Q4 2025 public disclosures of a major European G-SIB (publicly available Pillar 3 report). All actuals labelled as 2025A; forecast years are model-generated. Regulatory references are to publicly available EBA guidelines and EU regulations. No proprietary or confidential data used. Built for portfolio purposes only. Python projects consume the Master Balance Sheet CSV and extend the Excel analytical engine programmatically.